Day 29 – Stack Basics

📚 What is Stack?

Stack ek linear data structure hai jo LIFO (Last In First Out) principle follow karta hai.

👉 Matlab:

Jo element last me add hoga, wahi sabse pehle remove hoga

🧠 Real-life example:

Plate stack 🍽️

Browser history (back button)

🔑 Basic Operations of Stack

1. Push (Insert)

👉 Stack me element add karna

2. Pop (Delete)

👉 Top element remove karna

3. Peek / Top

👉 Top element dekhna (remove nahi hota)

4. isEmpty

👉 Stack empty hai ya nahi

📊 Stack Representation

Using Array

Stack: [10, 20, 30]
Top → 30

💻 Java Implementation (Using Array)

```java
package Day_29;

class StackArray {
    int top;
    int size;
    int arr[];

    StackArray(int size) {
        this.size = size;
        arr = new int[size];
        top = -1;
    }

    // Push
    void push(int x) {
        if (top == size - 1) {
            System.out.println("Stack Overflow");
            return;
        }
        arr[++top] = x;
        System.out.println(x + " pushed into stack");
    }

    // Pop
    int pop() {
        if (top == -1) {
            System.out.println("Stack Underflow");
            return -1;
        }
        return arr[top--];
    }

    // Peek
    int peek() {
        if (top == -1) {
            System.out.println("Stack is empty");
            return -1;
        }
        return arr[top];
    }

    // isEmpty
    boolean isEmpty() {
        return top == -1;
    }

    // Display Stack
    void display() {
        if (top == -1) {
            System.out.println("Stack is empty");
            return;
        }

        System.out.println("Stack elements:");
        for (int i = top; i >= 0; i--) {
            System.out.println(arr[i]);
        }
    }

    public static void main(String[] args) {

        StackArray stack = new StackArray(5);

        // Push elements
```

```java
        stack.push(10);
        stack.push(20);
        stack.push(30);
        stack.push(40);
        stack.push(50);

        // Try overflow
        stack.push(60);

        // Display
        stack.display();

        // Peek
        System.out.println("Top element: " + stack.peek());

        // Pop
        System.out.println("Popped: " + stack.pop());
        System.out.println("Popped: " + stack.pop());

        // Display after pop
        stack.display();

        // Check empty
        System.out.println("Is stack empty? " + stack.isEmpty());
    }
}
```

🔄 Dry Run

Push Operations:

push(10)
push(20)
push(30)

Stack:

[10, 20, 30]
Top = 30

Pop Operation:

pop() → 30

Stack:

[10, 20]
Top = 20

⚡ Important Points (Interview)

Stack follows LIFO

Push & Pop → O(1) time complexity

Overflow & Underflow conditions important hai

Implementation:

Array

Linked List

Built-in Stack (Java)

🚀 Java Built-in Stack (Shortcut)

```
import java.util.Stack;

class Main {
    public static void main(String[] args) {
        Stack<Integer> st = new Stack<>();

        st.push(10);
        st.push(20);
        st.push(30);

        System.out.println(st.pop());   // 30
        System.out.println(st.peek());  // 20
    }
}
```